



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

PQ-Shield: Security-Performance Benchmarking of PQC Algorithms for AI API Protection

Course / Review: B.Tech Project 1 — Review 1 | **School:** School of Computer Science and Engineering (SCOPE) |

Institution: Vellore Institute of Technology (VIT), Vellore

Project Team - 20768UG01

- **Shashwat Kumar** — Reg. No: 23BCE2080
- **Manav Prakash** — Reg. No: 23BCE0548
- **Yatharth Srivastava** — Reg. No: 23BCE0543

Faculty Guide

Dr. K. Nivitha

Assistant Professor Sr. Grade 1

School of Computer Science and Engineering (SCOPE)

VIT Vellore

The Harvest-Now-Decrypt-Later Threat to AI Infrastructure

Modern production AI inference endpoints — hosted on OpenAI, AWS SageMaker, or custom FastAPI/Flask servers — transmit sensitive diagnostic features, proprietary financial data, and confidential model logits. These payloads are currently protected by classical cryptographic schemes that are fundamentally vulnerable to quantum attack.

Growing AI API Reliance

Production AI inference endpoints transmit sensitive diagnostic features, proprietary financial data, and confidential logits. These payloads carry enormous commercial and strategic value, making them prime targets for sophisticated adversaries with long-term intelligence objectives. The volume of AI API traffic is accelerating rapidly, expanding the attack surface with every deployment.

Cryptographic Vulnerability

Existing endpoints rely on RSA-2048 key exchange and ECDSA signatures. Shor's algorithm, running on cryptographically relevant quantum computers (CRQCs), will break both schemes in polynomial time. The mathematical hardness assumptions underpinning these schemes — integer factorization and discrete logarithm problems — collapse entirely under quantum computation, rendering all currently intercepted traffic retroactively decryptable.

Harvest-Now-Decrypt-Later (HNDL)

Adversaries are passively capturing encrypted AI payload streams today to decrypt them retroactively once quantum hardware matures. State-level actors and well-resourced criminal organizations are already archiving TLS sessions at scale. The window for action is closing: data intercepted today may be decrypted within 5–10 years as CRQCs become operational.

The AI Inference SLA Gap

NIST standardized PQC algorithms in August 2024 (ML-KEM / FIPS 203 and ML-DSA / FIPS 204). However, no empirical studies measure how PQC migration impacts strict AI latency service level agreements (SLAs < 100 ms) under real-world concurrency. This gap leaves security architects without the data needed to make informed migration decisions for latency-sensitive AI workloads.

Project Aim, Abstract & Core Objectives

Project Aim

To design and implement PQ-Shield, an empirical measurement framework that evaluates the operational trade-offs like latency, byte overhead, CPU, and memory of NIST post-quantum cryptographic wrappers for real-time AI inference APIs.

Abstract

Artificial intelligence systems rely heavily on RESTful APIs to deliver real-time model inferences, yet these endpoints remain secured by classical RSA and ECDSA cryptography. This creates an urgent vulnerability to Harvest-Now-Decrypt-Later (HNDL) attacks, where adversaries log encrypted API traffic today to retroactively decrypt it once Cryptographically Relevant Quantum Computers (CRQCs) emerge. While migrating to Post-Quantum Cryptography (PQC) is essential, the operational overhead of integrating NIST-standardized algorithms—specifically ML-KEM (FIPS 203) and ML-DSA (FIPS 204)—into latency-sensitive AI inference pipelines under sub-100ms Service Level Agreements (SLAs) remains largely undocumented.

To address this gap, this project introduces PQ-Shield, an empirical benchmarking framework that evaluates quantum-resistant RESTful AI endpoints built on FastAPI. Using liboqs primitives, PQ-Shield evaluates three operational setups: Classical Baseline (RSA-2048 + ECDSA P-256), Pragmatic Hybrid (ML-KEM-768 + ECDSA P-256), and Pure Post-Quantum (ML-KEM-768 + ML-DSA-65). Under concurrency sweeps from 10 to 1,000 connections and simulated threat scenarios like HNDL byte expansion and MITM tamper detection, PQ-Shield measures handshake latency, round-trip time, memory bounds, and CPU utilization to generate a multi-axis trade-off matrix that guides security architects in balancing quantum defense with strict performance needs.

Core Objectives

1 Multi-Configuration Cryptographic Wrapping

Wrap a real-time FastAPI inference server in three configurations: **Classical** (RSA/ECDSA), **Hybrid** (ML-KEM + ECDSA), and **Full PQC** (ML-KEM + ML-DSA). Each configuration is implemented as a swappable middleware layer, enabling clean A/B/C comparisons under identical workload conditions.

2 Concurrency Benchmarking

Subject endpoints to load sweeps at **10, 100, and 1,000 concurrent connections** to measure handshake latency, round-trip time (RTT), CPU utilization, and memory bounds. Statistical rigor is ensured through 5 repetitions per cell with full distributional reporting (median, mean, standard deviation, p95, p99).

3 Adversarial Threat Simulation

Evaluate HNDL payload storage growth across 1,000 requests and test active Man-in-the-Middle (MITM) tamper-detection speeds. These scenarios integrate live passive data harvesting and active response tampering directly into the execution pipeline, providing realistic threat-contextualized measurements.

4 Decision Matrix Construction

Construct a weighted, multi-axis trade-off matrix to guide security architects in choosing migration strategies. The matrix balances security rating against normalized latency overhead, CPU cost, and memory footprint across all concurrency levels, replacing theoretical claims with empirical evidence.

Comparative Survey of PQC Integration & Benchmarking Studies

The following table synthesizes eight foundational studies examining PQC performance across heterogeneous computing environments, industry TLS deployments, Python production tooling, and next-generation network infrastructure. Collectively, they establish the empirical baseline against which PQ-Shield's novelty is positioned.

Paper / Reference	Objective	Methodology	Pros / Cons	Key Findings
Abbasi et al. (2025)	Benchmark NIST PQC in heterogeneous systems	Empirical execution across various CPU architectures	Pros: Broad hardware sweep. Cons: No AI workload or network stack.	ML-KEM demonstrates competitive keygen speeds vs RSA.
Demir et al. (2025)	Analyze PQC deployment in industry settings	TLS & telecom throughput benchmarking	Pros: Industry deployment focus. Cons: Lacks application-layer API metrics.	Hybrid key exchange adds negligible network overhead in bulk TLS.
Egbuagha & Ikwunna (2025)	Review practical PQC adoption post-NIST	Systematic literature review	Pros: Comprehensive theoretical summary. Cons: No experimental data.	Highlights severe gap in high-concurrency API measurements.
Pote & Bansode (2024)	Evaluate PQC algorithm efficiency	Micro-benchmarking core cryptographic ops	Pros: Detailed algorithm primitives. Cons: Isolated tests, no web protocol context.	Lattice operations show asymmetric sign/verify behavior.
Shaw (2026)	Bridge Python cryptography production gaps	Python library for X25519 + ML-KEM hybrid	Pros: Production Python implementation. Cons: No adversarial threat models.	Hybrid key exchange overhead is bounded in Python async runtimes.
Montenegro et al. (2025)	Framework for Post-Quantum TLS 1.3	Layered TLS handshake performance evaluation	Pros: Protocol-level accuracy. Cons: Restricted to generic TLS traffic.	PQC key size increases cause TCP packet fragmentation.
Chhetri et al. (2025)	Survey quantum-safe migration challenges	Qualitative risk & compliance analysis	Pros: Clear migration taxonomy. Cons: Lacks empirical performance numbers.	Hybrid approaches recommended for short-to-medium term transitions.
Faval & Moreira (2026)	Evaluate PQC in 5G core network security	Integration into open-source 5G testbeds	Pros: Real-world telecom application. Cons: Telecom-specific network profiles.	PQC verification latency fits within 5G control-plane requirements.



Research Gap Identified: No study combines AI-specific payload profiling, high-concurrency load sweeps, and adversarial threat simulation within a single empirical framework — the precise gap PQ-Shield is designed to fill.

Comparative Survey of Threat Modeling & Domain Benchmarks

The second set of seven studies extends the review into quantum-safe threat modeling for next-generation networks, ML endpoint privacy risks, HNDL economic feasibility, and PQC signature performance. These works collectively underscore the absence of integrated AI-API security benchmarks with adversarial context.

Paper / Reference	Objective	Methodology	Pros / Cons	Key Findings
Kudaloor & Aijaz (2026)	Quantum-safe evaluation for 6G networks	Experimental evaluation of PQC in wireless nodes	Pros: Edge-node testing. Cons: Does not cover REST/JSON API pipelines.	Lattice signatures introduce packet size overhead in low-MTU links.
Song et al. (2019)	Analyze privacy risks in ML endpoints	Membership inference under adversarial defense	Pros: ML security focus. Cons: Pre-dates practical PQC standards.	ML inference endpoints require payload integrity verification.
Tidjon & Khomh (2022)	Assess threat vectors in ML-based systems	Empirical threat mapping using MITRE ATLAS	Pros: Comprehensive attack surface. Cons: Ignores transport-layer cryptography.	API endpoints are vulnerable to payload interception and tampering.
Ahmed et al. (2025)	Survey PQC support in crypto libraries	Code analysis of OpenSSL, liboqs, and Botan	Pros: Highlights developer tooling. Cons: No end-to-end service measurement.	liboqs bindings provide the most stable path for PQC integration.
Blanco-Romero et al. (2026)	Feasibility of Harvest-Now-Decrypt-Later attacks	Economic modeling across TLS 1.3, QUIC, SSH	Pros: First quantitative HNDL cost model. Cons: No application-layer payload context.	Storage cost of harvested traffic is a major defensive lever.
Gómez-Cambronero et al. (2026)	Layered TLS 1.3 performance evaluation	High-load key exchange benchmarking (100 req/s)	Pros: Multi-layer latency isolation. Cons: Fixed key exchange; no signature sweeps.	Hybrid PQC retains classical handshake stability under medium load.
Schemitt et al. (2025)	Impact of PQC digital signatures on platforms	Benchmarking PQC signatures across levels 1–5	Pros: Rigorous signature comparison. Cons: Domain restricted to blockchain payloads.	ML-DSA verification speed outperforms conventional ECDSA schemes.



Synthesis: Across all 15 surveyed papers, a consistent pattern emerges — PQC benchmarking remains siloed by domain (TLS, 5G, blockchain) with no unified framework addressing AI API workloads under adversarial conditions. PQ-Shield directly addresses this fragmentation.

Existing Systems vs. PQ-Shield: Research Gap & Novelty

Existing Systems — Limitations

Classical Crypto (RSA/ECDSA)

Broken by Shor's Algorithm on CRQCs. High HNDL exposure with no quantum-resistant fallback. All currently intercepted traffic remains permanently vulnerable to retroactive decryption.

Generic TLS Benchmarks

Arbitrary static payload bytes that ignore asymmetric API shapes. No active threat testing. Results are not transferable to AI inference workloads with strict SLA budgets.

Proposed: PQ-Shield

Swappable Crypto Wrappers

Config A: RSA-2048 + ECDSA P-256 (Classical Baseline)
Config B: ML-KEM-768 + ECDSA P-256 (Hybrid)
Config C: ML-KEM-768 + ML-DSA-65 (Full PQC)

AI-Specific Workload Benchmark

Small requests (~0.5 KB) vs. larger responses (~0.2 KB) measured under real SLA budgets (<100 ms). Asymmetric AI payload profiles are explicitly modeled.

Active HNDL & MITM Simulation

Live passive data harvesting and active response tampering integrated directly into the execution pipeline, providing threat-contextualized measurements absent from prior work.

Three Dimensions of Project Novelty

AI Traffic Profile Specificity

Existing PQC benchmarks evaluate bulk web traffic. PQ-Shield measures PQC overhead on **asymmetric AI payloads** (small input vectors vs. dense logit responses) under tight SLAs (<100 ms), a profile entirely absent from the literature.

Adversarial Context Integration

Integrates live passive data harvesting (HNDL payload expansion) and active response tampering (MITM verification latency) directly into the execution pipeline — a combination not found in any of the 15 surveyed studies.

Empirical Decision Model

Replaces theoretical claims with an **automated trade-off matrix** balancing security rating against normalized latency overhead across multi-concurrency sweeps, providing actionable guidance for security architects.

Dataset & System Specifications

Inference Workload Specifications

Primary Inference Model: Scikit-Learn Random Forest Classifier (100 estimators) trained on the **UCI Optical Recognition of Handwritten Digits Dataset**.

- **Dataset Size:** 1,797 image samples
- **Feature Dimension:** 64 numerical features per sample (8×8 pixel matrix, pixel values 0–16)
- **Payload Profile:** Input request $\approx$ 0.5 KB JSON; Output response $\approx$ 0.2 KB JSON (10-class probabilities + class prediction)

Secondary Inference Model (Payload Expansion Test): PyTorch Convolutional Neural Network (CNN) trained on **CIFAR-10** (32×32×3 image tensor inputs) to validate overhead across larger feature arrays and stress-test PQC wrappers under heavier payload conditions.

Experimental Platform & Software Stack

Backend Server Framework

Python 3.11, FastAPI, Uvicorn ASGI server — chosen for native async support and minimal overhead in high-concurrency scenarios.

Classical Cryptography

Python cryptography library (OpenSSL backend) for RSA-2048 and ECDSA P-256 operations in Configurations A and B.

Post-Quantum Libraries

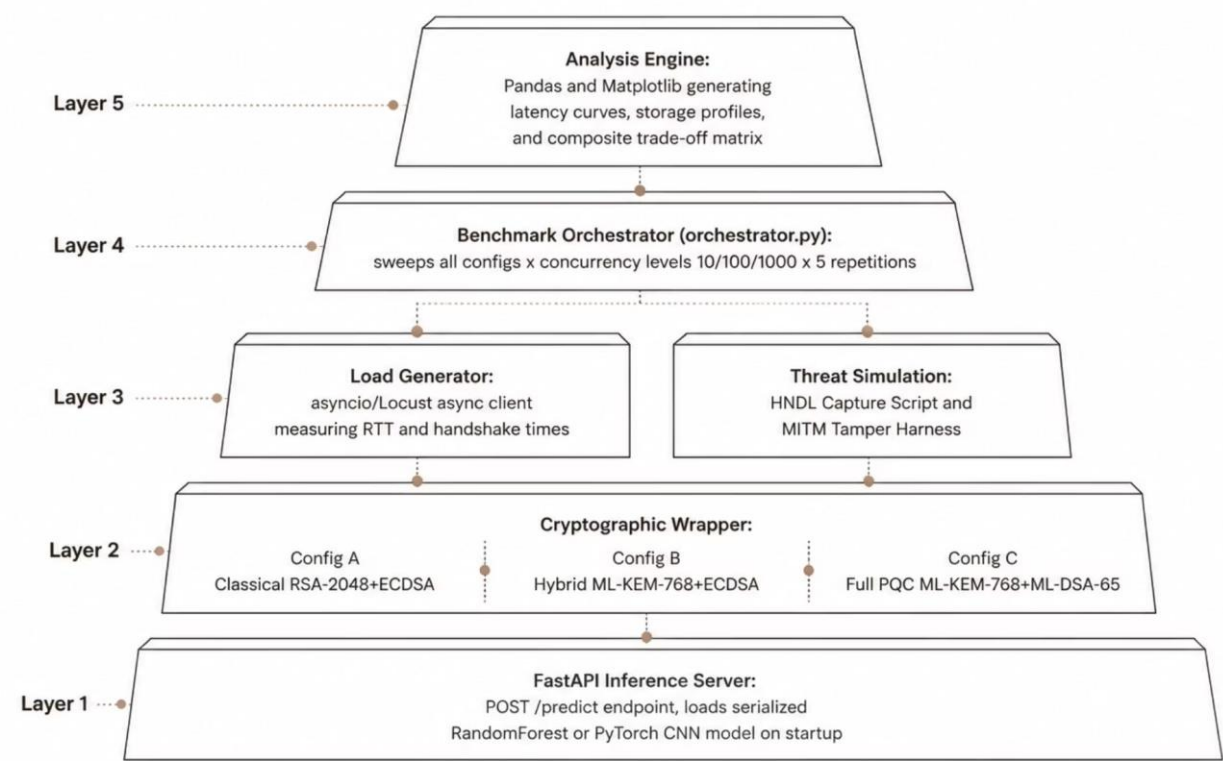
liboqs (Open Quantum Safe C library) with pyoqs Python bindings — the most stable and actively maintained PQC integration path per Ahmed et al. (2025).

Load & Threat Simulation

httpx (async client), mitmproxy (interception harness), Scapy / Raw Sockets (HNDL capture), psutil & tracemalloc (resource profiling).

Proposed System Architecture: Five-Layer Modular Benchmarking

PQ-Shield's architecture is organized into five discrete layers, each responsible for a specific function within the benchmarking pipeline. This modular design enables independent testing, clean substitution of cryptographic configurations, and reproducible experimentation across all threat scenarios.



Layer 1–2: Inference & Cryptographic Core

The **FastAPI Inference Server** (Layer 1) exposes the `POST /predict` endpoint, loading serialized model artifacts (RandomForest or PyTorch CNN) on startup. The **Cryptographic Wrapper** (Layer 2) implements three swappable configurations via FastAPI middleware and dependency injection: Config A (Classical: RSA-2048 + ECDSA P-256), Config B (Hybrid: ML-KEM-768 + ECDSA P-256), and Config C (Full PQC: ML-KEM-768 + ML-DSA-65). Each configuration is a drop-in replacement, enabling clean A/B/C comparisons under identical workload conditions.

Layer 3–5: Load, Threat, Orchestration & Analysis

The **Load Generator** (Layer 3, left) uses `asyncio` and `Locust` to generate async client traffic, measuring RTT and handshake times. The **Threat Simulation** module (Layer 3, right) runs HNDL capture scripts and MITM tamper harnesses in parallel. The **Benchmark Orchestrator** (Layer 4) executes the full sweep matrix: {Config A, B, C} × {Concurrency 10, 100, 1000} × {5 Repetitions}. The **Analysis Engine** (Layer 5) consumes raw CSV metrics to generate latency curves, storage profiles, and the composite trade-off matrix using `Pandas` and `Matplotlib`.

SDG Alignment, Project Outcomes & References

Alignment with UN Sustainable Development Goals



SDG 9: Industry, Innovation & Infrastructure

Target 9.c: Upgrade technological infrastructure to ensure enterprise AI resilience against post-quantum cyber threats. PQ-Shield provides the empirical foundation for quantum-safe AI infrastructure upgrades, directly supporting the modernization of critical digital systems.



SDG 16: Peace, Justice & Strong Institutions

Target 16.6: Build effective, accountable, and transparent digital security institutions protecting privacy. By quantifying PQC trade-offs for AI APIs, PQ-Shield enables institutions to make evidence-based security decisions that protect citizen and enterprise data.

Target Project Outcomes & Technology Readiness

→ Target Publication Venue

Submitting research findings to **IEEE Access** or an **ACM CCS Workshop**, targeting peer-reviewed dissemination of the empirical benchmarking methodology and trade-off matrix.

→ Open-Source Software Artifact

PQ-Shield measurement framework released on GitHub with full **Dockerized reproducibility**, enabling independent verification and extension by the research community.

→ Technology Readiness Level

Moving from **TRL 3** (Experimental Proof of Concept) to **TRL 4** (Technology Validated in Lab Environment) through systematic benchmarking across all three configurations and concurrency levels.

Key References

1. M. Abbasi, F. Cardoso, P. Vaz, J. Silva, and P. Martins, "A Practical Performance Benchmark of Post-Quantum Cryptography Across Heterogeneous Computing Environments," *Cryptography*, vol. 9, no. 1, p. 12, 2025.
2. M. Demir et al., "Performance Analysis and Industry Deployment of Post-Quantum Cryptography Algorithms," arXiv preprint arXiv:2503.12952, 2025.
3. O. Egbuagha and E. Ikwunna, "Post-Quantum Cryptography in Practice: A Literature Review," *IACR Cryptology ePrint Archive*, vol. 2025, p. 1668, 2025.
4. P. Pote and R. Bansode, "Performance Evaluation of Post-Quantum Cryptography," *Journal of Information Systems Engineering and Management*, vol. 9, no. 2, pp. 45–53, 2024.
5. A. Shaw, "Bridging the Post-Quantum Production Gap with a Hybrid-by-Default Python Cryptography Library," arXiv preprint arXiv:2605.17061, 2026.
6. J. A. Montenegro, R. Rios, and J. Lopez-Cerezo, "A Performance Evaluation Framework for Post-Quantum TLS," *Future Generation Computer Systems*, vol. 175, pp. 108–120, 2026.
7. G. Chhetri, S. Somvanshi, P. Hebli, S. Brotee, and S. Das, "Post-Quantum Cryptography and Quantum-Safe Security: A Comprehensive Survey," arXiv preprint arXiv:2510.10436, 2025.
8. R. A. Faval and R. Moreira, "Empowering Mobile Networks Security Resilience by using Post-Quantum Cryptography," arXiv preprint arXiv:2603.28626, 2026.
9. A. Kudaloor and A. Aijaz, "Toward Quantum-Safe 6G: Experimental Evaluation of Post-Quantum Cryptography Techniques," arXiv preprint arXiv:2605.06881, 2026.
10. L. N. Tidjon and F. Khomh, "Threat Assessment in Machine Learning Based Systems," arXiv preprint arXiv:2207.00091, 2022.
11. J. Blanco-Romero, F. A. Mendoza, C. G. Rubio, C. Campo, and D. D. Sanchez, "On the Practical Feasibility of Harvest-Now, Decrypt-Later Attacks," arXiv preprint arXiv:2603.01091, 2026.
12. D. Gomez-Cambronero, D. Munteanu, and A. I. Gonzalez-Tablas, "Layered Performance Analysis of TLS 1.3 Handshakes: Classical, Hybrid, and Pure Post-Quantum Key Exchange," arXiv preprint arXiv:2603.11006, 2026.